

Volume & Measure

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Converting Units

Every conversion is a power of ten.

Warm-Up 12 ITEMS • RECALL

1 1 m in cm

2 1 km in m

3 1 kg in g

4 2 m in cm

5 3 km in m

6 5 L in mL

7 Centimetres in a metre

8 Metres in a kilometre

9 Millimetres in a centimetre

10 Grams in a kilogram

11 Millilitres in a litre

12 Centimetres in 3 metres

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 250 cm in m

14. 1,500 g in kg

15. 0.75 km in m

16. 3,200 mL in L

17. 45 mm in cm

18. 2.5 m in cm

19. 4,500 m in km

20. 0.75 kg in g

21. 250 mL in L

22. 85 mm in cm



Working space



Error journal

Counting Cubes

Volume is a count before it is a formula.

Warm-Up 12 ITEMS • RECALL

1 A 2 by 2 by 2 cube —
volume

2 A 3 by 1 by 1 box

3 A 2 by 3 by 1 box

4 A 4 by 2 by 1 box

5 A 3 by 3 by 1 box

6 A 1 by 1 by 5 box

7 Cubes in a 2 by 2 by 2
box

8 In a 3 by 1 by 1 box

9 In a 2 by 3 by 1 box

10 In a 4 by 2 by 1 box

11 In a 3 by 3 by 1 box

12 In a 1 by 1 by 5 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4 by 3 by 2 box

.....

14. A 5 by 4 by 3 box

.....

15. One layer of a 5 by 4 box holds

.....

16. That box 3 layers high

.....

17. A 6 by 5 by 2 box

.....

18. Cubes in a 3 by 3 by 3 box

.....

19. In a 4 by 3 by 2 box

.....

20. In a 5 by 4 by 2 box

.....

21. In a 6 by 2 by 3 box

.....

22. In a 5 by 5 by 2 box

.....



!

Volume Formulas

Length $\times$ width $\times$ height, and a missing edge you have to find.

Warm-Up 12 ITEMS • RECALL

1 $2 \times 3 \times 4$

2 $5 \times 5 \times 2$

3 $10 \times 2 \times 3$

4 $1 \times 7 \times 4$

5 $6 \times 2 \times 2$

6 $3 \times 3 \times 3$

7 Volume of a 2 by 3 by 4 box

8 Of a 5 by 2 by 2 box

9 Of a 3 by 3 by 3 cube

10 Of a 1 by 6 by 2 box

11 Of a 4 by 4 by 1 box

12 Of a 2 by 2 by 5 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Volume 120, base area 20 — the height
.....14. Volume 100, height 5 — the base area
.....15. A 12 by 5 by 4 tank
.....16. Volume 72, edges 6 and 3 — the third edge
.....17. A cube of edge 4
.....18. Volume of a 6 by 5 by 4 box
.....19. Of a 10 by 3 by 3 box
.....20. Of a 8 by 4 by 2 box
.....21. Of a 7 by 5 by 3 box
.....22. A cube of side 4 — its volume
.....

★ Challenge NOT GRADED • REASON IT OUT

23. Two boxes: $4 \times 3 \times 2$ and $5 \times 2 \times 2$ — total volume

24. A cube of volume 125 — its edge

25. Of a 12 by 8 by 5 box

26. A cube of side 10 — its volume

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Line Plots

Fractional measurements, plotted and then reasoned about.

Warm-Up 12 ITEMS • RECALL

1 Four items at $\frac{1}{2}$ each — total

2 Eight items at $\frac{1}{4}$ each — total

3 Two items at $\frac{3}{4}$ each — total, as a/b

4 Three items at $\frac{1}{3}$ each

5 Six items at $\frac{1}{2}$ each

6 Longest of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{8}$ — type it

7 A line plot with 3 dots at $\frac{1}{2}$ — the count

8 Dots at $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$ — the total count

9 $\frac{1}{4} + \frac{1}{4}$ — type as a/b

10 $\frac{1}{2} + \frac{1}{2}$

11 Three dots at $\frac{1}{4}$ — their total as a/b

12 Two dots at $\frac{1}{2}$ — their total

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$ — total

14. Range of $\frac{1}{8}$ and $\frac{7}{8}$ — type as a/b

15. Five items totalling $\frac{5}{2}$ — the mean, as a/b

16. Difference between $\frac{7}{8}$ and $\frac{3}{8}$ — as a/b

17. Four measurements of $\frac{3}{4}$ — total

18. Dots at $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$ — the total

19. Four dots at $\frac{3}{4}$ — their total

20. The range of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — type as a/b

21. Dots at $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$ — the total

22. Six dots at $\frac{1}{3}$ — their total

★ Challenge NOT GRADED • REASON IT OUT

23. Six pencils totalling $9\frac{1}{2}$ inches, shared equally — each, as a/b

24. Two at $1/8$, three at $1/4$, one at $1/2$ — total, as a/b

25. Eight dots at $1/2$ shared evenly among 4 — each gets

26. Dots at $1/4$, $1/2$, $1/2$, $3/4$ — the mean as a/b

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Box It

Enrichment. Same volume, different shapes.

✦ Warm-Up 4 ITEMS • RECALL

- 1 A box of volume 24: 2 by 3 by ?

- 2 A box of volume 24: 1 by 4 by ?

- 3 A 2 by 3 by 4 box — volume

- 4 A cube of side 3

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A box of volume 36: 3 by 3 by ?
.....
6. A box of volume 60: 5 by 4 by ?
.....
7. A 6 by 4 by 5 box
.....
8. A box of volume 60 with base 12 — its height
.....

Challenge NOT GRADED • REASON IT OUT

9. How many whole boxes of volume 8 fit in one of volume 96

10. A cube with the same volume as a $2 \times 4 \times 8$ box — its edge

11. Double every edge of a $2 \times 3 \times 4$ box. Volume multiplies by

12. A 12 by 5 by 4 box

13. Which holds more — type 10x2x6 or 5x4x6

14. A cube of volume 125 — its side

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Converting Units

OUT OF 22

Warm-Up • 1-12 • 100 | 1000 | 1000 | 200 | 3000 | 5000 | 100 | 1000 | 10 | 1000 | 1000 | 300

Core • 13-22 • 2.5 | 1.5 | 750 | 3.2 | 4.5 | 250 | 4.5 | 750 | 0.25 | 8.5

Challenge • 23-26 • 2500000 | 205 | 120000 | 3.5

1.2 • Counting Cubes

OUT OF 22

Warm-Up • 1-12 • 8 | 3 | 6 | 8 | 9 | 5 | 8 | 3 | 6 | 8 | 9 | 5

Core • 13-22 • 24 | 60 | 20 | 60 | 60 | 27 | 24 | 40 | 36 | 50

Challenge • 23-26 • 320 | 4 | 120 | 240

1.3 • Volume Formulas

OUT OF 22

Warm-Up • 1-12 • 24 | 50 | 60 | 28 | 24 | 27 | 24 | 20 | 27 | 12 | 16 | 20

Core • 13-22 • 6 | 20 | 240 | 4 | 64 | 120 | 90 | 64 | 105 | 64

Challenge • 23-26 • 44 | 5 | 480 | 1000

1.4 • Line Plots

OUT OF 22

Warm-Up • 1-12 • 2 | 2 | $\frac{3}{2}$ | 1 | 3 | $\frac{1}{2}$ | 3 | 3 | $\frac{1}{2}$ | 1 | $\frac{3}{4}$ | 1

Core • 13-22 • 1 | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 3 | $\frac{3}{2}$ | 3 | $\frac{1}{2}$ | $\frac{3}{4}$ | 2

Challenge • 23-26 • $\frac{3}{4}$ | $\frac{3}{2}$ | 1 | $\frac{1}{2}$

Fri • Box It

OUT OF 8

Warm-Up • 1-4 • 4 | 6 | 24 | 27

Core • 5-8 • 4 | 3 | 120 | 5

Challenge • 9-14 • 12 | 4 | 8 | 240 | same | 5

Count the Cubes

Build it, count it, write it. The formula comes later.

Warm-Up 12 ITEMS • RECALL

1 A 2 by 2 by 2 cube

2 A 3 by 1 by 1 box

3 A 2 by 3 by 1 box

4 A 4 by 2 by 1 box

5 A 3 by 3 by 1 box

6 A 1 by 1 by 5 box

7 Cubes in a 2 by 2 by 3 box

8 In a 4 by 1 by 1 box

9 In a 3 by 2 by 2 box

10 In a 5 by 2 by 1 box

11 In a 4 by 4 by 1 box

12 In a 1 by 1 by 8 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4 by 3 by 2 box

.....

14. A 5 by 4 by 3 box

.....

15. One layer of a 5 by 4 box

.....

16. That box 3 layers high

.....

17. A 6 by 5 by 2 box

.....

18. Cubes in a 4 by 4 by 4 box

.....

19. In a 5 by 3 by 2 box

.....

20. In a 6 by 4 by 2 box

.....

21. In a 7 by 3 by 3 box

.....

22. In a 6 by 6 by 2 box

.....



Working space



Error journal

Layers

One layer, times the height. That is the whole idea.

Warm-Up 8 ITEMS • RECALL

1 A 3 by 4 base — cubes in one layer

2 Two such layers

3 A 5 by 2 base — one layer

4 Three such layers

5 A 6 by 1 base — one layer

6 Four such layers

7 A 2 by 5 base — one layer

8 A 4 by 4 base — one layer

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. A 7 by 3 base, 4 high

.....

10. A 8 by 5 base, 2 high

.....

11. Base area 20, height 6

.....

12. Base area 15, height 4

.....

13. Volume 100, base area 20 — the height

.....

14. A 6 by 5 base, 4 layers

.....

15. A 7 by 3 base, 5 layers

.....

16. A 8 by 4 base, 3 layers

.....

17. A volume of 90 with a 9 by 2 base — layers

.....

18. A volume of 120 with a 10 by 4 base — layers

.....



The Formula

Length $\times$ width $\times$ height, and where each one comes from.

Warm-Up 12 ITEMS • RECALL

1 $2 \times 3 \times 4$

2 $5 \times 5 \times 2$

3 $10 \times 2 \times 3$

4 $1 \times 7 \times 4$

5 $6 \times 2 \times 2$

6 $3 \times 3 \times 3$

7 Volume of a 3 by 4 by 2 box

8 Of a 5 by 5 by 2 box

9 Of a 6 by 2 by 2 box

10 Of a 4 by 3 by 3 box

11 A cube of side 2

12 Of a 7 by 2 by 1 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 12 by 5 by 4 tank
.....

14. A cube of edge 4
.....

15. A 9 by 3 by 2 box
.....

16. A cube of edge 5
.....

17. A 20 by 10 by 5 box
.....

18. Volume of a 9 by 4 by 3 box
.....

19. Of a 8 by 6 by 2 box
.....

20. Of a 11 by 3 by 4 box
.....

21. A cube of side 6
.....

22. Of a 10 by 10 by 2 box
.....

★ Challenge NOT GRADED • REASON IT OUT

23. A cube of volume 125 — its edge

24. A cube of volume 216 — its edge

25. Of a 15 by 8 by 6 box

26. A cube of side 12

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Find a Missing Edge

Given volume and two edges, work out the third.

Warm-Up 12 ITEMS • RECALL

1 Volume 12, edges 2 and 3 — the third

2 Volume 24, edges 2 and 3

3 Volume 30, edges 5 and 3

4 Volume 8, edges 2 and 2

5 Volume 20, edges 5 and 2

6 Volume 36, edges 6 and 3

7 Volume 24, base 12 — the height

8 Volume 30, base 10 — the height

9 Volume 36, base 9 — the height

10 Volume 20, base 4 — the height

11 Volume 48, base 24 — the height

12 Volume 18, base 6 — the height

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Volume 72, edges 6 and 3

.....

14. Volume 120, edges 5 and 4

.....

15. Volume 240, edges 12 and 5

.....

16. Volume 100, height 5 — the base area

.....

17. Volume 96, edges 8 and 4

.....

18. Volume 120, base 20 — the height

.....

19. Volume 144, length 6 width 4 — the height

.....

20. Volume 210, length 7 height 5 — the width

.....

21. Volume 96, width 4 height 3 — the length

.....

22. A cube of volume 64 — its side

.....

★ Challenge NOT GRADED • REASON IT OUT

23. Volume 360, edges 9 and 5

.....

24. Volume 1000 with equal edges — one edge

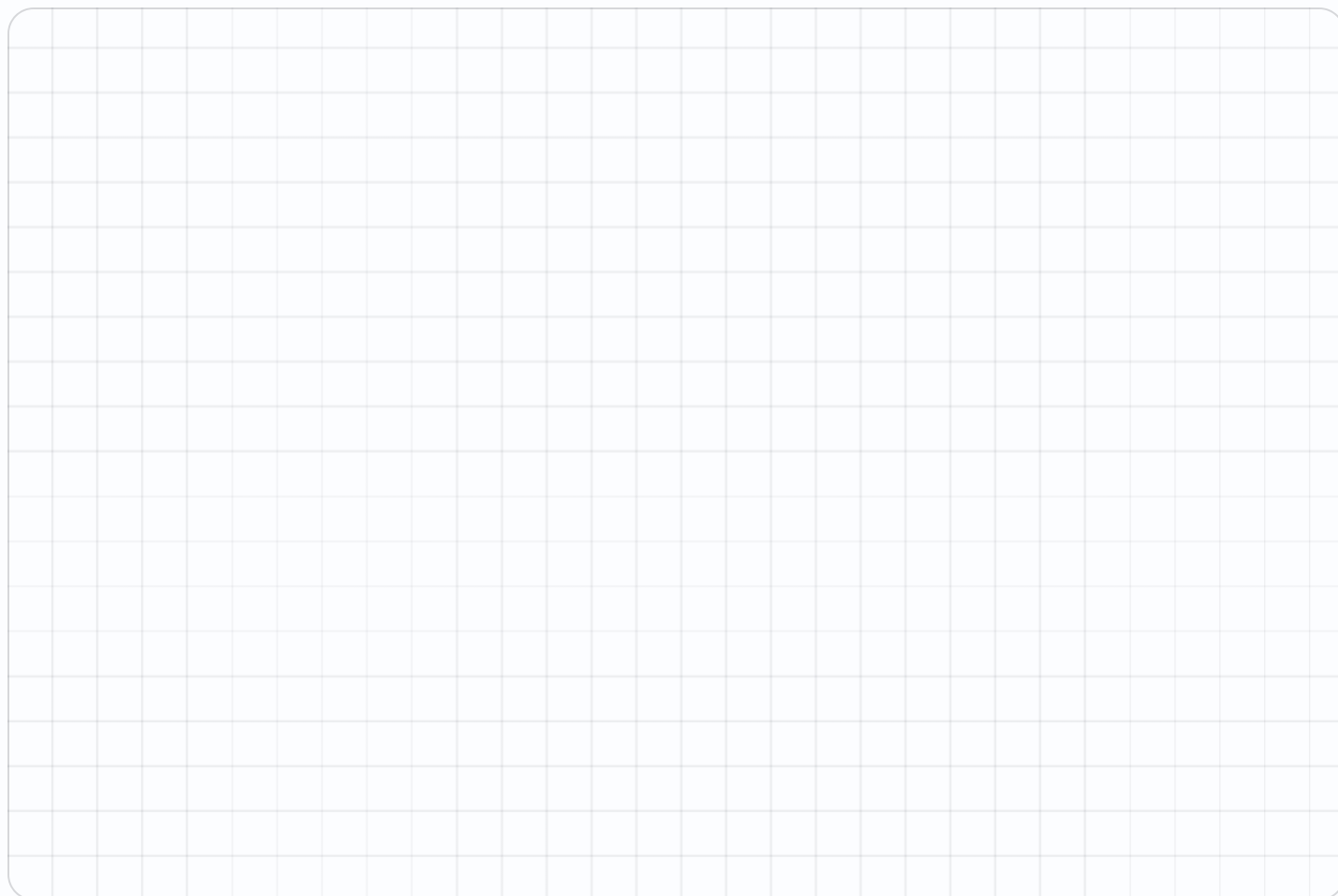
.....

25. Volume 720, length 12 width 6 — the height

.....

26. A cube of volume 343 — its side

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Cube Count

Read the drawing, call the volume before your opponent does.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

+

Warm-Up

4 ITEMS • RECALL

1 A 2 by 2 by 3 box

2 A 3 by 3 by 2 box

3 A 2 by 2 by 2 cube — volume

4 A 3 by 3 by 3 cube

◆

Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A 4 by 4 by 4 cube

6. A 5 by 3 by 4 box

7. A 5 by 4 by 3 box

8. Volume 60, base 20 — the height



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Count the Cubes

OUT OF 22

Warm-Up • 1-12 • 8 | 3 | 6 | 8 | 9 | 5 | 12 | 4 | 12 | 10 | 16 | 8

Core • 13-22 • 24 | 60 | 20 | 60 | 60 | 64 | 30 | 48 | 63 | 72

Challenge • 23-26 • 320 | 4 | 150 | 240

2.2 • Layers

OUT OF 18

Warm-Up • 1-8 • 12 | 24 | 10 | 30 | 6 | 24 | 10 | 16

Core • 9-18 • 84 | 80 | 120 | 60 | 5 | 120 | 105 | 96 | 5 | 3

Challenge • 19-22 • 6 | 360 | 504 | 7

2.3 • The Formula

OUT OF 22

Warm-Up • 1-12 • 24 | 50 | 60 | 28 | 24 | 27 | 24 | 50 | 24 | 36 | 8 | 14

Core • 13-22 • 240 | 64 | 54 | 125 | 1000 | 108 | 96 | 132 | 216 | 200

Challenge • 23-26 • 5 | 6 | 720 | 1728

2.4 • Find a Missing Edge

OUT OF 22

Warm-Up • 1-12 • 2 | 4 | 2 | 2 | 2 | 2 | 2 | 3 | 4 | 5 | 2 | 3

Core • 13-22 • 4 | 6 | 4 | 20 | 3 | 6 | 6 | 6 | 8 | 4

Challenge • 23-26 • 8 | 10 | 10 | 7

Fri • Cube Count

OUT OF 8

Warm-Up • 1-4 • 12 | 18 | 8 | 27

Core • 5-8 • 64 | 60 | 60 | 3

Challenge • 9-14 • 44 | 8 | 216 | 240 | 6 | 64

Split the Solid

Draw the dividing line first, then measure each part.

Warm-Up 11 ITEMS • RECALL

1 A 2 by 2 by 2 plus a 2 by 2 by 1

2 A 3 by 2 by 1 plus a 3 by 2 by 1

3 A 4 by 2 by 2 plus a 2 by 2 by 2

4 A 5 by 1 by 1 plus a 3 by 1 by 1

5 Parts in an L-shaped solid

6 A 2 by 2 by 5 box

7 An L split into a 2 by 3 by 1 and a 2 by 2 by 1 — the first volume

8 The second volume

9 Their total

10 A 3 by 3 by 1 plus a 3 by 1 by 1

11 A 5 by 2 by 1 plus a 5 by 1 by 1

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. A $6 \times 4 \times 2$ plus a $3 \times 4 \times 2$

13. A $10 \times 5 \times 2$ minus a $4 \times 5 \times 2$

14. An L of $8 \times 3 \times 2$ and $4 \times 3 \times 2$

15. A $5 \times 5 \times 4$ minus a $2 \times 2 \times 4$

16. A T of $6 \times 2 \times 2$ and $2 \times 4 \times 2$

17. A 6 by 4 by 2 plus a 4 by 2 by 2

18. A 10 by 5 by 2 plus a 5 by 5 by 2

19. A 8 by 3 by 3 with a 2 by 3 by 3 removed

20. A 6 by 6 by 3 with a 2 by 2 by 3 removed

21. A 12 by 4 by 2 plus a 4 by 4 by 2

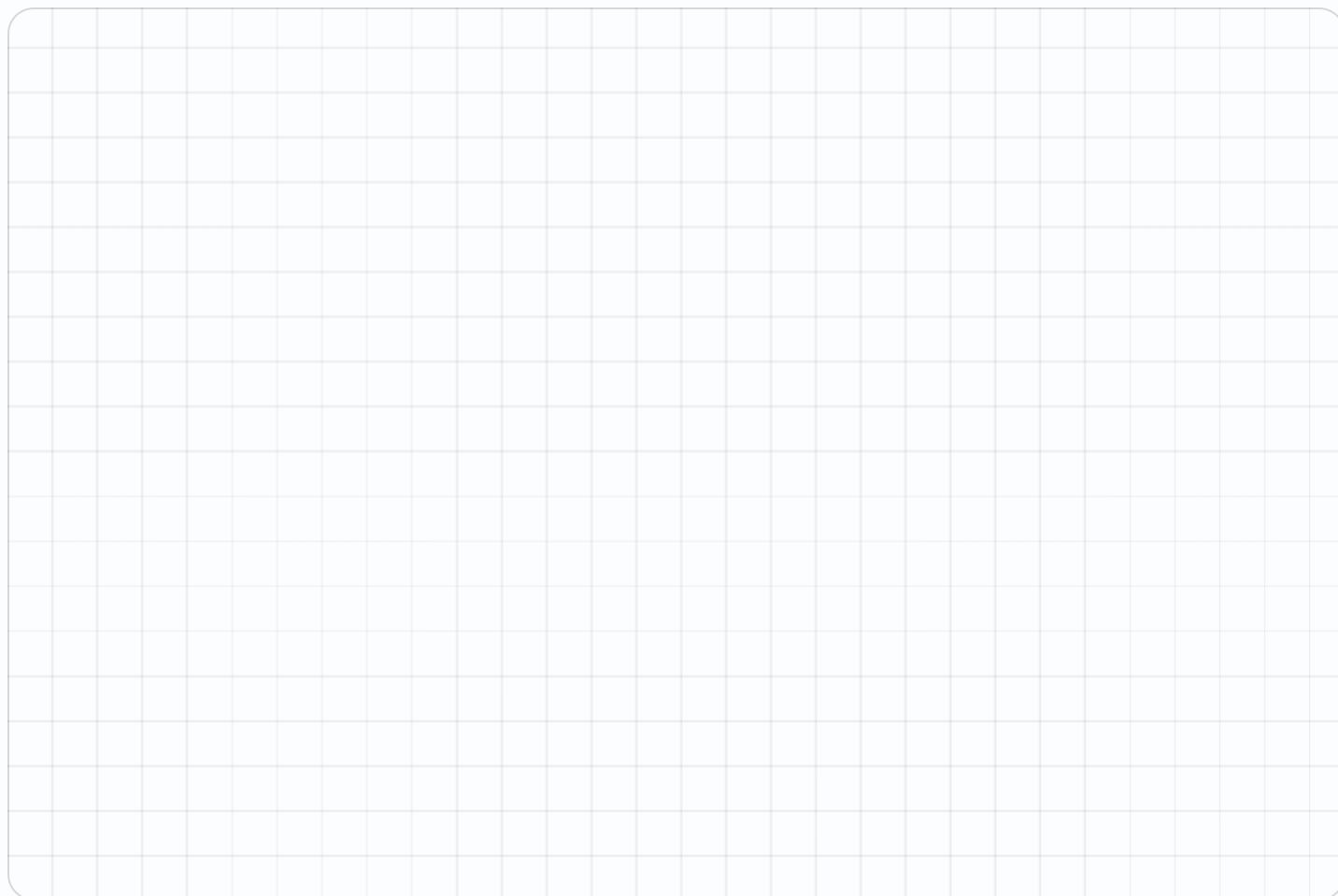
★ Challenge NOT GRADED • REASON IT OUT

22. A $10 \times 10 \times 5$ with a $4 \times 4 \times 5$ hole

23. A $12 \times 6 \times 3$ plus a $6 \times 6 \times 3$

24. A 10 by 10 by 4 with a 5 by 4 by 4 removed

25. A staircase of three 4 by 2 by 1 steps

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Add the Volumes

Two boxes, one total. Check each part before adding.

Warm-Up 9 ITEMS • RECALL

1 $12 + 8$

2 $24 + 16$

3 A $2 \times 2 \times 2$ and a $3 \times 2 \times 2$ —
total

4 A $4 \times 2 \times 1$ and a $4 \times 2 \times 1$

5 $30 + 45$

6 A $5 \times 2 \times 2$ and a $5 \times 2 \times 2$

7 $18 + 27$

8 $50 + 50$

9 $36 + 24$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. A $6 \times 5 \times 2$ and a $4 \times 3 \times 2$

11. A $8 \times 4 \times 3$ and a $2 \times 2 \times 3$

12. A $10 \times 3 \times 2$ and a $5 \times 3 \times 2$

13. Three $4 \times 3 \times 2$ boxes

14. A $7 \times 4 \times 2$ and a $3 \times 4 \times 2$

15. A 5 by 4 by 3 plus a 4 by 3 by 2

16. A 7 by 5 by 2 plus a 6 by 3 by 2

17. A 9 by 4 by 3 plus a 3 by 3 by 3

18. A 8 by 5 by 4 plus a 5 by 4 by 2

19. A 10 by 6 by 2 minus a 4 by 3 by 2

★ Challenge NOT GRADED • REASON IT OUT

20. A $12 \times 8 \times 4$ and a $6 \times 4 \times 4$

21. Five $6 \times 5 \times 2$ boxes

22. A 12 by 8 by 5 minus a 6 by 4 by 5

23. Three boxes of 4 by 3 by 2 stacked

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Line Plots

Plot measurements to the nearest eighth, then reason about them.

Warm-Up 12 ITEMS • RECALL

1 Four items at $\frac{1}{2}$ — total

2 Eight items at $\frac{1}{4}$ — total

3 Three items at $\frac{1}{3}$ — total

4 Six items at $\frac{1}{2}$ — total

5 Two items at $\frac{3}{4}$ — total

6 Longest of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{8}$

7 Dots at $\frac{1}{2}$, $\frac{1}{2}$ — the total

8 Dots at $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{4}$

9 Three dots at $\frac{1}{2}$ — the total as a/b

10 Two dots at $\frac{3}{4}$

11 Four dots at $\frac{1}{8}$ — the total as a/b

12 Five dots at $\frac{1}{5}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$

.....

14. Range of $\frac{1}{8}$ and $\frac{7}{8}$

.....

15. Five items totalling $\frac{5}{2}$ — the mean

.....

16. Difference between $\frac{7}{8}$ and $\frac{3}{8}$

.....

17. Four measurements of $\frac{3}{4}$ — total

.....

18. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total

.....

19. Dots at $\frac{1}{8}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ — the total

.....

20. Six dots at $\frac{1}{4}$ shared among 3 — each gets as a/b

.....

21. The range of $\frac{1}{8}$ to $\frac{7}{8}$ — type as a/b

.....

22. Dots at $\frac{1}{3}$, $\frac{2}{3}$, 1 — the total

.....

★ Challenge NOT GRADED • REASON IT OUT

23. Six pencils totalling $9/2$ inches shared equally — each

24. Two at $1/8$, three at $1/4$, one at $1/2$ — total

25. Dots at $1/2$, $1/2$, $1/4$, $3/4$ shared among 4 — each as a/b

26. Eight dots at $3/8$ — their total

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Read the Plot

Total, difference, and redistribution.

Warm-Up 11 ITEMS • RECALL

1 Five values of $\frac{1}{2}$ — total
_____2 Their mean
_____3 Four values of $\frac{1}{4}$ — total
_____4 Range of $\frac{1}{4}$ and $\frac{3}{4}$
_____5 Range of $\frac{1}{8}$ and $\frac{5}{8}$
_____6 The most common value on a plot is the — type mode or mean
_____7 Dots at $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{4}$ — the mode as a/b
_____8 The biggest minus the smallest is the
_____9 The range of $\frac{1}{4}$ and $\frac{3}{4}$ — type as a/b
_____10 Dots at $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$ — the mode
_____11 How many dots make 4 measurements

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Values $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — total
.....13. Values $\frac{1}{8}$, $\frac{3}{8}$, $\frac{1}{2}$ — total
.....14. Four values totalling 3 — the mean
.....15. Range of $\frac{1}{8}$ and 1
.....16. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total
.....17. That plot's mean, shared among 4 — type as a/b
.....18. That plot's range as a/b
.....19. Dots at $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$ — the total as a/b
.....20. Dots at $\frac{2}{3}$, $\frac{2}{3}$, $\frac{1}{3}$ — the total as a/b
.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 1 m in cm

2 A 2 by 3 by 4 box

3 A 3 by 4 by 2 box —
volume

4 Centimetres in a metre

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 250 cm in m

.....

6. 1500 g in kg

.....

7. A 5 by 4 by 3 box

.....

8. Volume 120, base area 20 — the height

.....

9. Four measurements of $\frac{3}{4}$ — total

.....

10. A 6 by 5 by 4 box

.....

11. Volume 96, base 24 — the height

.....

12. 2.5 m in cm

.....

13. A 5 by 4 by 3 plus a 4 by 3 by 2

.....

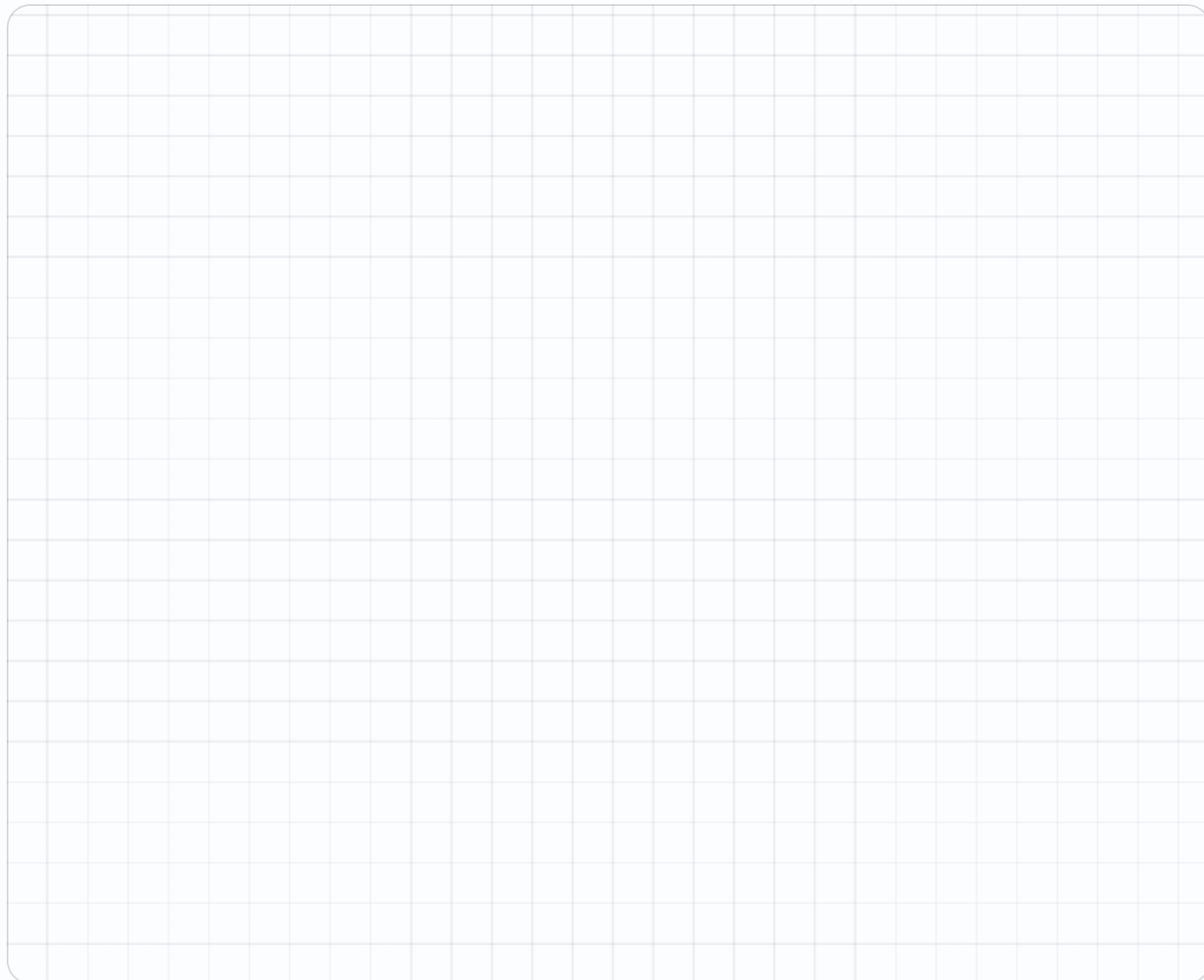
14. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total as a/b

.....

★ Challenge NOT GRADED • REASON IT OUT

15. A $10 \times 10 \times 5$ with a $4 \times 4 \times 5$ hole

16. A 10 by 10 by 4 with a 5 by 4 by 4 removed

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Split the Solid

OUT OF 21

Warm-Up • 1-11 • 12 | 12 | 24 | 8 | 2 | 20 | 6 | 4 | 10 | 12 | 15

Core • 12-21 • 72 | 60 | 72 | 84 | 40 | 64 | 150 | 54 | 96 | 128

Challenge • 22-25 • 420 | 324 | 320 | 24

3.2 • Add the Volumes

OUT OF 19

Warm-Up • 1-9 • 20 | 40 | 20 | 16 | 75 | 40 | 45 | 100 | 60

Core • 10-19 • 84 | 108 | 90 | 72 | 80 | 84 | 106 | 135 | 200 | 96

Challenge • 20-23 • 480 | 300 | 360 | 72

3.3 • Line Plots

OUT OF 22

Warm-Up • 1-12 • 2 | 2 | 1 | 3 | $\frac{3}{2}$ | $\frac{1}{2}$ | 1 | 1 | $\frac{3}{2}$ | $\frac{3}{2}$ | $\frac{1}{2}$ | 1

Core • 13-22 • 1 | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 3 | $\frac{3}{2}$ | 1 | $\frac{1}{2}$ | $\frac{3}{4}$ | 2

Challenge • 23-26 • $\frac{3}{4}$ | $\frac{3}{2}$ | $\frac{1}{2}$ | 3

3.4 • Read the Plot

OUT OF 20

Warm-Up • 1-11 • $\frac{5}{2}$ | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | mode | $\frac{1}{2}$ | range | $\frac{1}{2}$ | $\frac{1}{2}$ | 4

Core • 12-20 • $\frac{3}{2}$ | 1 | $\frac{3}{4}$ | $\frac{7}{8}$ | 2 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{5}{4}$ | $\frac{5}{3}$

Challenge • 21-24 • $\frac{5}{8}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 100 | 24 | 24 | 100

Core • 5-14 • 2.5 | 1.5 | 60 | 6 | 3 | 120 | 4 | 250 | 84 | $\frac{3}{2}$

Challenge • 15-16 • 420 | 320

Design Three Boxes

Same volume, different shapes. All three must check out.

Warm-Up 12 ITEMS • RECALL

1 A box of volume 24: 2 by 3 by ?

2 Volume 24: 1 by 4 by ?

3 Volume 24: 2 by 2 by ?

4 Volume 12: 2 by 3 by ?

5 Volume 36: 3 by 3 by ?

6 Volume 60: 5 by 4 by ?

7 A 2 by 3 by 4 box — volume

8 A 1 by 4 by 6 box

9 A 2 by 2 by 6 box

10 A 1 by 2 by 12 box

11 A 3 by 8 by 1 box

12 A 2 by 12 by 1 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Volume 48: 4 by 4 by ?
.....

14. Volume 72: 6 by 3 by ?
.....

15. Volume 96: 8 by 4 by ?
.....

16. Volume 120: 6 by 5 by ?
.....

17. Volume 64 with equal edges — one edge
.....

18. Three boxes all of volume 36 — one is 3 by 3 by 4, another 2 by 3 by 6, the third 1 by 6 by 6. How many
.....

19. A 36-volume box with a 9 base — its height
.....

20. A 36-volume box with a 12 base — its height
.....

21. A 48-volume box with a 4 by 4 base — its height
.....

22. A cube nearest to volume 64 — its side
.....

23. A cube with the same volume as a $2 \times 4 \times 8$ box — its edge
24. How many whole boxes of volume 8 fit in one of volume 96
25. Whole-number boxes of volume 12 with distinct dimensions — how many, counting $1 \times 1 \times 12$, $1 \times 2 \times 6$, $1 \times 3 \times 4$, $2 \times 2 \times 3$
26. A 100-volume box with a 25 base — its height

Argue for One

Which shape is best, and best for what?

Warm-Up 12 ITEMS • RECALL

1 A $2 \times 2 \times 6$ box — volume

2 A $2 \times 3 \times 4$ box — volume

3 A $1 \times 4 \times 6$ box — volume

4 Are those volumes equal — yes or no

5 A 24 cm^3 box on a 12 cm^2 base — its height

6 A cube of edge 3 — volume

7 Volume of a 2 by 5 by 5 box

8 Of a 5 by 5 by 2 box

9 Same volume — type yes or no

10 A tall thin box of 1 by 2 by 25

11 Of a 10 by 5 by 1 box

12 Of a 25 by 2 by 1 box

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Surface area of a $2 \times 2 \times 6$ box

.....

14. Surface area of a $2 \times 3 \times 4$ box

.....

15. Which uses less card — type $2 \times 2 \times 6$ or $2 \times 3 \times 4$

.....

16. Surface area of a cube of edge 3

.....

17. Surface area of a $1 \times 4 \times 6$ box

.....

18. Which shape wastes least material for the same volume — type cube or long thin

.....

19. A 4 by 4 by 4 box — volume

.....

20. A 1 by 8 by 8 box — volume

.....

21. A 2 by 4 by 8 box — volume

.....

22. How many of those three hold the same

.....



Working space



Error journal

Conversions and Volume

Both strands together, which is how they arrive in real problems.

Warm-Up 12 ITEMS • RECALL

1 1 m in cm

2 1 kg in g

3 2 m in cm

4 3 L in mL

5 A 2×2×2 box

6 500 g in kg

7 Centimetres in a metre

8 Millilitres in a litre

9 Grams in a kilogram

10 A 10 by 10 by 10 cm box
— cubic cm

11 That box in litres

12 Millimetres in a
centimetre

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 250 cm in m

14. 1500 g in kg

15. A 100 by 50 by 20 cm box — volume in cubic cm

16. 3200 mL in L

17. A 2 by 1 by 0.5 m tank — volume in cubic m

18. A 20 by 10 by 10 cm box — cubic cm

19. A 2 L container in cubic cm

20. A 500 mL bottle in cubic cm

21. A 30 by 20 by 10 cm tank in litres



Working space



Error journal

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 1 km in m

2 A 3 by 3 by 3 cube

3 1 L in mL

4 A 4 by 2 by 1 box

5 2 kg in g

6 Four items at $\frac{1}{2}$ — total

7 A 3 by 4 by 2 box — volume

8 A cube of side 3

9 Centimetres in 4 metres

10 Volume 30, base 10 — the height

11 Dots at $\frac{1}{2}$, $\frac{1}{2}$ — the total

12 2 L in mL

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 45 mm in cm

14. A 5 by 4 by 3 box

15. Volume 72, edges 6 and 3 — the third

16. 0.75 km in m

17. Range of $\frac{1}{8}$ and $\frac{7}{8}$

18. A 6 by 5 by 4 box

19. Volume 144, length 6 width 4 — the height

20. 0.75 kg in g

21. A 5 by 4 by 3 plus a 4 by 3 by 2

22. Dots at $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ — the total as a/b



Mission 07 Test

Twelve items plus the Big Question, answered out loud.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

◆ Warm-Up

4 ITEMS • RECALL

1 1 m in cm

2 A 2 by 3 by 4 box

3 A 2 by 3 by 5 box — volume

4 A cube of side 4

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 250 cm in m

6. 3200 mL in L

7. A 5 by 4 by 3 box

8. A cube of edge 4

9. Volume 120, base area 20 — the height

10. Volume 96, edges 8 and 4 — the third

11. A 6×5×2 and a 4×3×2 — total

12. Five items totalling 5/2 — the mean

13. A 7 by 4 by 3 box

14. Volume 120, base 24 — the height

15. 3.5 m in cm

16. 4,500 g in kg

17. A 6 by 4 by 2 plus a 4 by 2 by 2

18. Dots at 1/8, 1/4, 3/8 — the total as a/b

19. A cube of volume 125 — its side

20. A 20 by 10 by 10 cm box in litres



Working space



Error journal

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Design Three Boxes

OUT OF 22

Warm-Up • 1-12 • 4 | 6 | 6 | 2 | 4 | 3 | 24 | 24 | 24 | 24 | 24 | 24

Core • 13-22 • 3 | 4 | 3 | 4 | 4 | 3 | 4 | 3 | 3 | 4

Challenge • 23-26 • 4 | 12 | 4 | 4

4.2 • Argue for One

OUT OF 22

Warm-Up • 1-12 • 24 | 24 | 24 | yes | 2 | 27 | 50 | 50 | yes | 50 | 50 | 50

Core • 13-22 • 56 | 52 | $2 \times 3 \times 4$ | 54 | 68 | cube | 64 | 64 | 64 | 3

Challenge • 23-26 • cube | 96 | 6 | yes

4.3 • Conversions and Volume

OUT OF 21

Warm-Up • 1-12 • 100 | 1000 | 200 | 3000 | 8 | 0.5 | 100 | 1000 | 1000 | 1000 | 1 | 10

Core • 13-21 • 2.5 | 1.5 | 100000 | 3.2 | 1 | 2000 | 2000 | 500 | 6

Challenge • 22-25 • 1 | 1000 | 1000 | 60

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 1000 | 27 | 1000 | 8 | 2000 | 2 | 24 | 27 | 400 | 3 | 1 | 2000

Core • 13-22 • 4.5 | 60 | 4 | 750 | $\frac{3}{4}$ | 120 | 6 | 750 | 84 | $\frac{3}{2}$

Challenge • 23-26 • 6 | 205 | 720 | 6

Fri • Mission 07 Test

OUT OF 20

Warm-Up • 1-4 • 100 | 24 | 30 | 64

Core • 5-20 • 2.5 | 3.2 | 60 | 64 | 6 | 3 | 84 | $\frac{1}{2}$ | 84 | 5 | 350 | 4.5 | 64 | $\frac{3}{4}$ | 5 | 2

Challenge • 21-24 • 5 | 8 | 360 | 1000